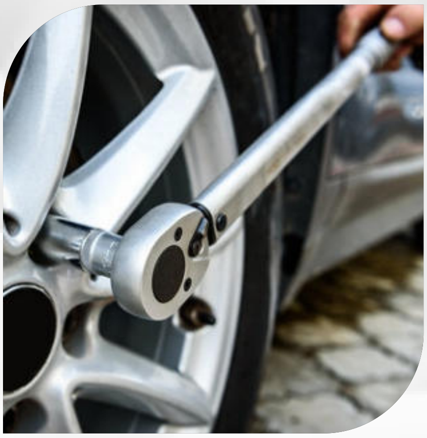
UNIT 04

TURNING EFFECT OF FORCES





PARALLEL FORCES

We often come across objects on which many forces are acting. In many cases, we find all or some of the forces acting on a body in the same direction.

For example

Many people push a bus to start it. Why do all of them push it in the same direction? All these forces are applied in the same direction so these are all parallel to each other. Such forces which are parallel to each other are called parallel forces.

LIKE AND UNLIKE PARALLEL FORCES

LIKE PARALLEL FORCES

Like parallel forces are the forces that are parallel to each other and have the same direction.

UNLIKE PARALLEL FORCES

Unlike parallel forces are the forces that are parallel but have directions opposite to Each other.

ADDITION OF FORCES

Force is a vector quantity. It has both magnitude and direction. It is represented bya line segment with an arrow-head at one end to show its direction of action. Length of the line segment gives the magnitude of the force on suitable scale. Wherever more than one force act on an object we need to add them to get a single resultant force.

RESULTANT FORCE

Single force that has the same effect as the combined effect of the forces to be added is called resultant force.

Ordinary arithmetic rules cannot be used to add the forces.

Two different methods are used for the addition of forces:

- **Graphical Method**
- **Analytical Method**

GRAPHICAL METHOD

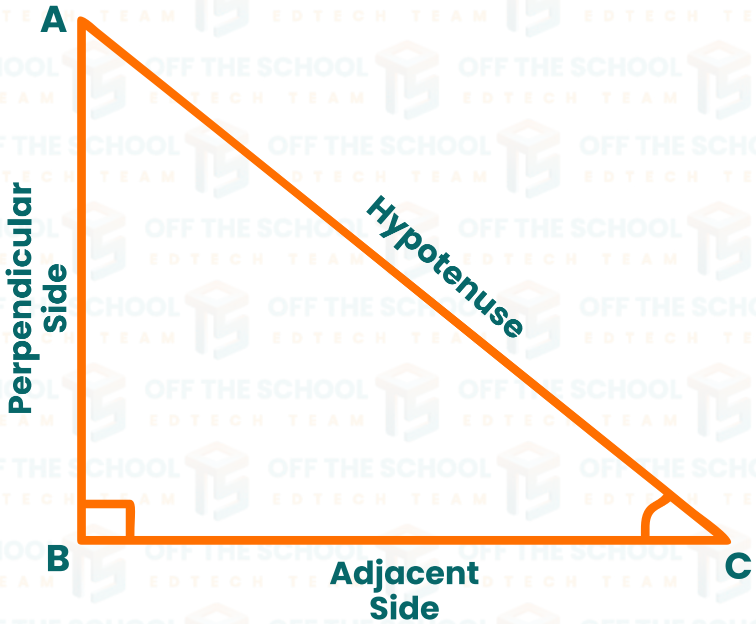
In this method forces can be added by **head to tail rule** of vector addition.

HEAD TO TAIL RULE

First select a suitable scale. Then draw the vectors of all the forces according to the scale: such as vectors **A** and **B**. Take any one of the vectors as first vector e.g. vector **A**. Then draw next vector **B** such that its tail coincides with the head of the first vector **A**. similarly draw the next vector for the third force (if any) with its tail coinciding with the head of the previous vector and so on. Now draw a vector **R** such that its tail is at the tail of vector **A**, the first vector, while its head is at the head of vector **B**, the last vector. Vector **R** represents the resultant force completely in magnitude and direction.

TRIGONOMETRIC RATIOS

The ratio between any two sides of a right angled triangle are given specific names. There are six ratios in total out of which three are main ratios and other three are their reciprocals. Three main ratios mostly used in physics are sine, cosine, and tangent. Consider a right-angled triangle △ABC having angle ϴ at C.



$$\text{Sin}\Theta = \frac{Perpendicular}{Hypotenuse} = \frac{AB}{AC}$$

$$\text{Cos}\Theta = \frac{Base}{Hypotenuse} = \frac{BC}{AC}$$

$$\tan\Theta = \frac{Perpendicular}{Base} = \frac{AB}{BC}$$

RESOLUTION OF FORCES

The process of splitting of a vector into mutually perpendicular components is called resolution of vectors. Consider a force F represented by a line segment OA which makes an angle with x-axis. Draw a perpendicular line AB on x-axis from A. The components OB = $\mathbf{F_x}$ and BA = $\mathbf{F_y}$ are perpendicular to each other. They are called the perpendicular components of OA= F.

Therefore,

$F = F_x + F_y$ ____________________ (i)

The trigonometric ratios can be used to find the magnitude $\mathbf{F_x}$ and $\mathbf{F_y}$. In right angled triangle △OBA.

$$\frac{F_x}{F} = \frac{OB}{OA} = \cos\Theta$$

$$F_x = F\cos\Theta \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ (ii)$$

$$\frac{F_y}{F} = \frac{BA}{OA} = \sin\Theta$$

$$F_y = F\sin\Theta \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ (ii)$$

Equation (ii) and (iii) give the perpendicular components respectively.

DETERMINATION OF FORCE FROM ITS PERPENDICULAR COMPONENTS

This is opposite to the process of resolution. If the perpendicular components of a force are known then the process of determining the force itself from the perpendicular components is called composition. Suppose **F_x** and **F_y** are the perpendicular components of the force F and are represented by line segments OP and PR with arrowhead respectively, Applying the head to tail rule:

$$OR = OP + PR$$

Here OR represents the force F whose x and y components are F_x and F_y respectively. Thus,

$$F = F_x + F_y$$

In order to find the magnitude of F applying Pyhtagorous theorem to right angled triangle △OPR;

$$(OR)^2 = (OP)^2 + (PR)^2$$

OR

$$F^2 = {F_x}^2 + {F_y}^2$$

Therefore,

$$F = \sqrt{F_x^2 + F_y^2}$$

The direction of force F with x-axis is given by:

$$\tan\Theta = \frac{PR}{OP} = \frac{F_y}{F_x}$$

$$\Theta = \tan^{-1}\frac{F_y}{F_x}$$

TORQUE

"The turning effect of force is called the moment of force or Torque"
It depends upon: The magnitude of force. The perpendicular distance of the point of application of force from the pivot or fulcrum. Moment of force about
a point = Force × Perpendicular distance from the point depending of their direction.

$$\tau = F \times d$$

It S.I unit is N-m.

LINE OF ACTION OF FORCE

The line along which a force acts is called the line of action of the force.

MOMENT ARM

The perpendicular distance between the axis of rotation and the line of action of the force is called the moment arm of the force. Moment are described as **clockwise** or **anticlockwise**.

PRINCIPLE OF MOMENT

A force that turns a spanner in the clockwise direction is generally used to tighten a nut such type of moment produced is called **clockwise moment**. A force is applied such that it turns the nut in the anticlockwise direction is generally loosen a nut such type of moment produced is called **anticlockwise moment**. A body initially at rest does not rotate if sum of all the clockwise moments acting on it is balanced by the sum of all the anticlockwise moments acting on it. This is known as the principle of moments. According to the principle of moments:
"A body is balanced if the sum of clockwise moments acting on the body is equal to the sum of anticlockwise moments acting on it".
It depends upon the following
1. The magnitude of force.
2. The perpendicular distance of the point of application of force from the pivot or fulcrum.

CENTRE OF MASS

Centre of mass of a system is such a point where an applied force causes the system to move with rotation. Or It is a point at which whole mass of the body is assumed to be concentrated.

CENTRE OF GRAVITY

It is a point where the whole weight of the body is assumed to be concentrated.

CENTRE OF GRAVITY OF SOME REGULAR SHAPED OBJECTS

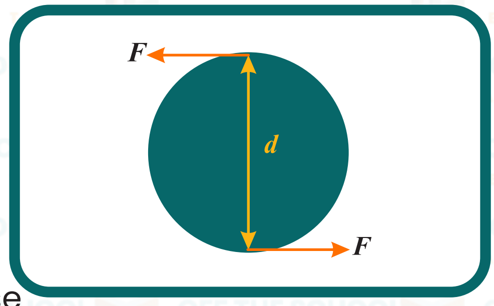
- The Center of gravity of regular-shaped uniform objects is their geometrical Center.
- The Center of gravity of a uniform rod is its midpoint.
- The Center of gravity of a uniform square or rectangular sheet is the point of intersection of its Diagonals.
- The Center of gravity of a solid or hollow sphere is the Center of the sphere.
- The Center of gravity of a uniform circular ring is the Center of the ring.
- The Center of gravity of a uniform circular disc is its Centre.
- The Center of gravity of a uniform solid or hollow cylinder is the mid-point on its axis.
- The Center of gravity of a uniform triangular sheet is the point of intersection of its medians.

CENTRE OF GRAVITY OF SOME IRREGULAR SHAPED OBJECTS

The Center of gravity of irregular-shaped thin lamina.

Step 1: Make three small holes near the edges of the lamina farther apart from each other.
Step 2: Suspend the lamina freely from one whole on retort stand through a pin.
Step 3: Hang a plumb line or weight from the pin in front of the lamina.
Step 4: When the plumb line is steady, trace the line on the lamina.
Step 5: Repeat steps 2 to 4 for second and third hole. The point of intersection of three lines is the position of center of gravity.

COUPLE



"Two unlike parallel forces of the same magnitude but not acting along the same line form a Couple".

For example

The forces required to turn the steering wheel of a car. The two equal and opposite forces balance, so the revolution will not move up, down, or sideways. However, if the wheel is not in equilibrium the pair of forces will cause it to rotate. A couple has a turning effect but does not cause an object to accelerate.

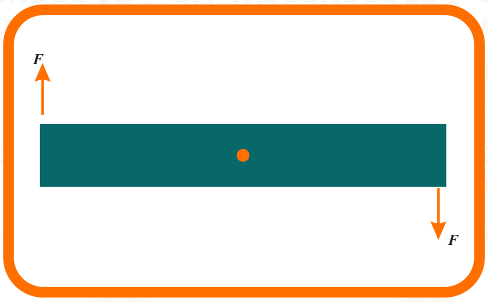
To form a couple, two forces must be:

- Equal in magnitude.
- Parallel, but opposite in direction.
- Separated by a distance.

We can calculate the torque of the couple in the above figure by adding the moments of each force about the Center O of the wheel:

Torque of couple = (F x OP) + (F x OQ)
Torque of couple = F x (OP+OQ)
Torque of couple = F x d
Torque of couple = One of the forces x Perpendicular distance between the forces.

EQUILIBRIUM



A body is said to be in Equilibrium when it is in or moving with uniform velocity.

OR

A body is said to be in equilibrium if no net force acts on it.

TYPES OF EQUILIBRIUM

There are two types of Equilibrium.

Static Equilibrium.

Dynamic Equilibrium.

STATIC EQUILIBRIUM

A body at rest is said to be in static equilibrium.

For example

- An object hanging at rest from a string.
- A book lying on a table.

DYNAMIC EQUILIBRIUM

A body is said to be in dynamic equilibrium when it is moving with uniform velocity.

For example

- A uniform downward motion of steel ball through viscous liquid.
- Jumping of paratrooper from the Helicopter.

CONDITIONS OF EQUILIBRIUM

There are two conditions for equilibrium.

FIRST CONDITION FOR EQUILIBRIUM

According to this condition for equilibrium sum of the all forces acting on a body must be equal to zero.

Suppose n number of forces $F_1, F_2, F_3, \ldots, F_n$ are acting on a body then according to first condition of the equilibrium:

$$F_1 + F_2 + F_3 + \ldots\ldots\ldots\ldots + F_n = 0$$

OR

$$\Sigma F = 0$$

In term of x and y components of the force acting on the body first condition for the equilibrium can be expressed as:

$$F_{1x} + F_{2x} + F_{3x} + \ldots\ldots\ldots\ldots + F_{nx} = 0 \text{ and}$$

$$F_{1y} + F_{2y} + F_{3y} + \ldots\ldots\ldots\ldots + F_{ny} = 0$$

OR

$$\Sigma F_x = 0$$

$$\Sigma F_y = 0$$

STATES OF EQUILIBRIUM

There are three states of Equilibrium:

Static equilibrium
Unstable equilibrium
Dynamic equilibrium

STATIC EQUILIBRIUM

The body is said to be in stable equilibrium if when slightly displaced and then released it returns to its previous position.

UNSTABLE EQUILIBRIUM

The body is said to be in unstable equilibrium when slightly tilted and does not return back to its previous position.

DYNAMIC EQUILIBRIUM

The body is said to be in neutral equilibrium when displaced from the previous position and remains in equilibrium in the new position.

Body is in stable equilibrium when	Body is in unstable equilibrium when	Body is said to be in neutral equilibrium when:
• Its Centre of gravity is at its lowest position.	• Its Centre of gravity is at the highest position.	• Its Center of gravity always remains above the point of contact.
• When it is tilted its Centre of gravity rises.	• When it is tilted its Centre of gravity is lowered.	• When it is displaced from its previous position it's Centre of gravity remains at the same height.
• It returns back to a stable state by lowering its Centre of gravity.	• Its previous position cannot be restored by raising.	• All the new states in which the body is moved are stable states.

STABILITY

Stability refers to the ability of an object or system to remain balanced or unchanged in its position, state, or behavior over time.

Explanation

In most situations, we are interested in maintaining stable equilibrium, or balance. For example, the Design of structures, racing cars, and in working with the human body. Consider a refrigerator if it is tilted slightly it will return back to its original position due to torque on it. But if it is tilted more it will fall down. The critical point is reached when the center of gravity shifts from one side of the pivot point to the other. When the center of gravity is on one side of the pivot point, the torque pulls the refrigerator back onto its original base of support If the refrigerator is tilted further, the center of gravity crosses onto the other side of the pivot point and The torque causes the refrigerator to topple.